A Two-Phase Integer L-Shaped Method for Stochastic Electric-Bus Wireless Charging Infrastructure Planning

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Abstract

This abstract will be written after the main body is stable.

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1. Introduction

Electrifying bus fleets requires coordinated planning of charging infrastructure, battery capacity, and the use of limited charging resources under uncertain future service requirements. Wireless charging technologies expand the set of charging opportunities by allowing buses to receive energy at selected stops or road links during regular service. This capability turns infrastructure planning from a stand-alone siting decision into a joint planning problem linking wireless facility deployment, route-level battery configuration, and the use of existing plug-in chargers.

Wireless charging facilities are long-lived assets, whereas transit service plans may change with future demand and procurement planning. A layout that is attractive for one future service plan may perform poorly under another because route frequencies, trip chains, active-fleet requirements, and charging-resource use may change. The infrastructure decision must therefore be made before the future service plan is known, whereas route-level battery configurations and operational decisions can adapt after a service-plan scenario is realized.

The central planning trade-off is shaped by the interaction among wireless infrastructure, battery capacity, and finite plug-in charging capacity. More in-route wireless charging can reduce battery requirements on some routes and lower their residual demand for plug-in charging. At the same time, facility placement changes when and where that residual demand occurs. Because plug-in chargers are limited shared resources, concurrent charging requirements couple route-level decisions that would otherwise be separable.

We capture this timing and coupling structure with a two-stage stochastic mixed-integer planning model. The first stage selects stationary and dynamic wireless charging facilities. The second stage selects route-level battery configurations and, conditional on those selections, determines representative-day charging allocation and energy management for the realized service-plan scenario. The resulting recourse problems contain discrete battery choices, plug-in charger-occupancy decisions, and shared plug-in capacity constraints. Consequently, both the deterministic equivalent and the classical integer L-shaped decomposition become difficult to scale as the network and scenario set grow.

To solve this model, we develop a reformulation-enabled integer L-shaped method. The method differs from a standard integer L-shaped decomposition by changing what recourse information is exposed to the master and by separating inexpensive lower-bound generation from exact recourse verification. This design preserves exactness when the cut loop converges and creates the algorithmic components evaluated in the computational study.

The paper makes three contributions. First, it identifies a layered two-stage stochastic mixed-integer structure in which here-and-now facility decisions interact with scenario-dependent discrete configurations and shared-resource operational recourse. The wireless-charging application provides a concrete instantiation in which conditioning on battery choices reveals route-local operations, limited plug-in capacity couples routes, and wireless facilities and batteries form a joint operational resource order. Second, it develops an exact reformulation-enabled integer L-shaped algorithm that combines scenario-dependent decision variables lifting, a two-phase relaxation-verification scheme, and monotonicity-strengthened cuts for mixed-integer recourse. Third, it provides comprehensive computational evidence through an extensive algorithmic ablation study and systematic analyses of performance with respect to network size, scenario count, and the gap between the Phase 1 lower-bounding relaxation and the exact Phase 2 recourse evaluation.

The experiments use wireless-charging planning instances constructed from Singapore bus-service data for routes operated by Tower Transit. They examine whether the model produces interpretable planning decisions,

how the algorithmic components affect network-scale and scenario-scale performance, and how the complete method responds as shared plug-in capacity makes the Phase 1 relaxation more or less aligned with exact verification.

The remainder of the paper is organized as follows. Section 2 reviews related work on electric-bus charging infrastructure planning and stochastic decomposition methods. Section 3 presents the two-stage formulation and identifies its recourse structure. Section 4 develops the lifted two-phase method and monotonicity-strengthened cuts. Section 5 reports the computational results. Section 6 concludes.

2. Literature Review

This paper intersects two literature streams. The first studies charging infrastructure, battery configuration, and energy-feasible operations for electric buses; the second develops exact decomposition methods for stochastic design problems with mixed-integer recourse.

2.1. Electric-bus charging and wireless infrastructure planning

Electric-bus planning links strategic infrastructure decisions with tactical battery and charging decisions. Charging-site selection and battery capacity determine the resources available to a fleet, while vehicle assignment, charging schedules, and route energy constraints determine whether those resources can support the prescribed service. Integrated optimization models therefore evaluate infrastructure and battery investments together with energy-feasible service operations.

Depot and terminal charging studies focus on the timing of plug-in charging under vehicle schedules, electricity prices, and charger-capacity limits. Opportunity-charging models extend this interaction to charging during service, allowing energy to be replenished between trips or at selected stops. When chargers have finite capacity, station–time occupancy constraints couple buses or routes that would otherwise admit separate energy-feasibility calculations.

Wireless charging introduces a further infrastructure layer. SWC creates charging opportunities at selected stations, whereas DWC supplies energy while a bus traverses an equipped road segment. The associated planning literature examines how the location and extent of these facilities affect battery requirements, route energy feasibility, and investment and charging costs. These models establish the facility–battery trade-off that underlies the present application.

Together, these studies provide the main application components of our model: wireless-facility placement, route-level battery configuration, energy-feasible trip chains, and limited plug-in charging resources. We combine these components under service-plan uncertainty, with wireless facilities selected before the future service plan is known and battery configurations and charging decisions adapted to its realization. This setting links long-term infrastructure planning to shared-resource mixed-integer recourse.

2.2. Decomposition methods for stochastic design problems

Two-stage stochastic models for facility location and network design separate infrastructure choices made before uncertainty from scenario-dependent recourse. Scenario decomposition is particularly attractive when the deterministic equivalent is large, because fixing the first-stage design separates the evaluation of distinct scenarios. The computational challenge increases when the recourse contains discrete configuration or operating decisions rather than a continuous linear program.

Benders decomposition and the L-shaped method use subproblem information to iteratively refine a master representation of future cost and feasibility. For continuous recourse, linear-programming duality supplies this information directly. Mixed-integer recourse instead requires integer L-shaped, logic-based, or related combinatorial mechanisms, and exact scenario evaluation can become the dominant computational burden.

One line of research improves this process by changing the decomposition architecture. Richer master formulations expose selected recourse decisions or partial recourse descriptions, while lower-bounding relaxations and internal subproblem decompositions reduce the cost of scenario evaluation. These choices determine both the strength of the master bound and the computational effort required to generate it.

A complementary line of research strengthens the information returned by the subproblems. Logic-based and combinatorial cuts exploit discrete feasibility structure, while dominance relations and problem-specific valid inequalities can extend a correction beyond the single configuration at which it was generated. Such strengthening depends on a verifiable ordering or implication structure in the underlying recourse model.

These studies show that master representation, subproblem organization, and cut design must be matched to the recourse structure. The setting considered here combines scenario-dependent discrete battery configurations, conditionally route-local operations, and limited shared-resource coupling within each scenario. This layered structure motivates an exact decomposition method designed around their interaction rather than a direct application of scenario decomposition alone.

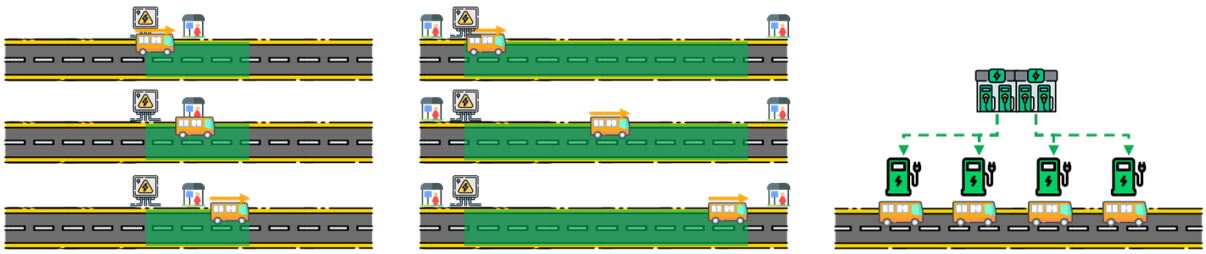
3. Problem Formulation and Recourse Structure

This section formulates the wireless charging infrastructure planning problem as a two-stage stochastic program. The formulation separates long-lived wireless-facility decisions from scenario-adaptive battery and charging decisions, and then identifies the recourse structure used by the algorithm in Section 4.

3.1. Two-stage stochastic formulation

Let Ω be the set of service-plan scenarios, and let p_ω be the probability of scenario $\omega \in \Omega$, where $p_\omega > 0$ and $\sum_{\omega\in\Omega} p_\omega = 1$. Each scenario represents one plausible future service plan and fixes the route frequencies, timetables, bus trip chains, active-fleet requirements, and trip-level energy requirements, while the operated route set $\mathcal{R}$ is common across scenarios. The planner first installs long-lived wireless charging facilities before the future service plan is known. After a scenario is realized, the second-stage recourse selects route-level battery capacities and determines the corresponding representative-day charging allocation and energy management.

Let $\mathcal{S}$ be the station set, $\mathcal{L}$ the directed link set between consecutive stations, and $C \subseteq \mathcal{S}$ the set of existing interchanges or hubs equipped with shared plug-in chargers. Stationary wireless charging (SWC) can be installed at any station in $\mathcal{S} \setminus C$, while dynamic wireless charging (DWC) can be installed on any directed link in $\mathcal{L}$. The first-stage binary variables w_s^{S} and w_l^{D} indicate whether SWC is installed at station $s \in \mathcal{S} \setminus C$ and DWC is installed on link $l \in \mathcal{L}$, respectively. Stations in C have fixed, limited plug-in capacity and are not candidates for SWC deployment. Figure 1 illustrates the three charging resources represented in the formulation.



(a) SWC **(b)** DWC **(c)** Plug-in charging

Figure 1: Charging resources represented in the formulation. (a) SWC is installed near selected stations and supplies energy during low-speed station approaches and departures. (b) DWC is installed on selected directed links and supplies energy during link traversals. (c) Existing plug-in chargers at hubs and interchanges provide limited station–time capacity shared across routes.

The compact two-stage stochastic formulation is

$$\min_{w^{\mathrm{S}},w^{\mathrm{D}}} \quad C^{\mathrm{fac}}(w) + \sum_{\omega\in\Omega} p_\omega \Phi_\omega(w) \tag{1}$$

$$\text{s.t.} \quad w_s^{\mathrm{S}} \in \{0,1\}, \qquad s \in \mathcal{S} \setminus C, \tag{2}$$

$$w_l^{\mathrm{D}} \in \{0,1\}, \qquad l \in \mathcal{L}. \tag{3}$$

where $w = (w^{\mathrm{S}}, w^{\mathrm{D}})$ and

$$C^{\mathrm{fac}}(w) = \sum_{s\in\mathcal{S}\setminus C} c_s^{\mathrm{S}} w_s^{\mathrm{S}} + \sum_{l\in\mathcal{L}} c_l^{\mathrm{D}} w_l^{\mathrm{D}}.$$

Here c_s^{S} and c_l^{D} denote the equivalent daily infrastructure costs, obtained by annualizing the installation costs over the planning horizon and converting them to a representative-day basis. Battery investment costs are converted on the same basis, so that all objective terms are expressed on a common daily scale. The vector w is the nonanticipative here-and-now facility decision, and $\Phi_\omega(w)$ is the optimal second-stage value in scenario ω.

Let z_ω collect the route-level battery selections, x_ω the plug-in charger occupancy decisions, and y_ω the charging amounts and battery energy states. For a fixed facility layout w, the scenario recourse value $\Phi_\omega(w)$ is the minimum scenario cost

$$\Phi_\omega(w) = \min_{z_\omega,x_\omega,y_\omega} \left\{ C_\omega^{\mathrm{bat}}(z_\omega) + C_\omega^{\mathrm{chg}}(y_\omega) : (z_\omega, x_\omega, y_\omega) \in \mathcal{P}_\omega(w) \right\}. \tag{4}$$

Fleet-size adjustment costs and the residual value of the existing batteries are scenario constants and are therefore omitted from $\Phi_\omega(w)$. The next subsection expands $\mathcal{P}_\omega(w)$.

3.2. Scenario recourse formulation

For a fixed scenario ω and route $r \in \mathcal{R}$, let $\mathcal{M}_{\omega,r}$ contain the bus trip chains on route r. Each chain corresponds to one bus duty consisting of an ordered sequence of trips, so $n_{\omega,r} = |\mathcal{M}_{\omega,r}|$ is the route fleet requirement in scenario ω. The sets $\mathcal{V}_{\omega,r}$ and $\mathcal{A}_{\omega,r}$ collect, respectively, all station-visit and directed-traversal event copies generated by these trip chains. Each event copy uniquely identifies its trip chain, trip, and position within the duty, so explicit event-ownership indices are suppressed for notational simplicity. Distinct operational occurrences remain separate even when they correspond to the same physical station or directed link.

Figure 2 illustrates the event-level representation of a representative bus duty and the charging opportunities embedded in its sequence of trips.

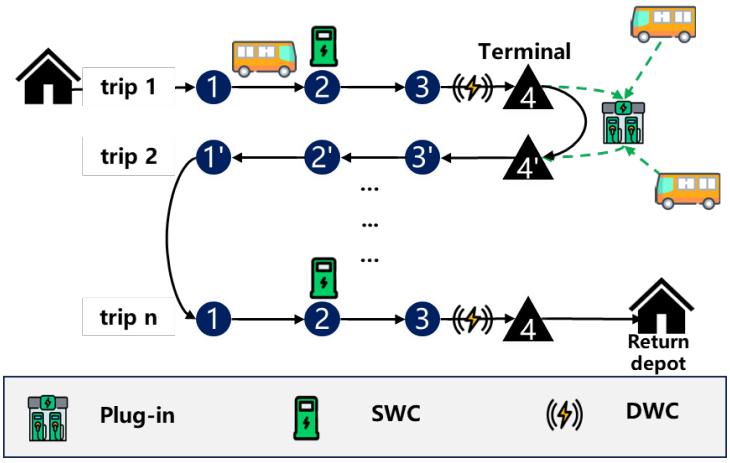


Figure 2: Event-level representation of a representative bus duty and its charging opportunities. A bus sequentially operates multiple trips composed of ordered station visits and directed traversals. SWC opportunities are associated with eligible station visits, DWC opportunities with equipped traversals, and plug-in charging with eligible terminal or hub visits. Limited plug-in chargers may be shared by buses arriving from different trips or routes.

A consolidated notation table is provided in Appendix Appendix A.

Battery configuration and cost.. Battery option $k \in \mathcal{K}$ has capacity β_k, and the scenario-dependent binary variable $z_{\omega,r,k}$ indicates whether it is selected for route r. The route-level battery-choice constraints are

$$\sum_{k\in\mathcal{K}} z_{\omega,r,k} = 1, \qquad r \in \mathcal{R}, \tag{5}$$

$$B_{\omega,r} = \sum_{k\in\mathcal{K}} \beta_k z_{\omega,r,k}, \qquad r \in \mathcal{R}, \tag{6}$$

$$z_{\omega,r,k} \in \{0,1\}, \qquad r \in \mathcal{R},\ k \in \mathcal{K}. \tag{7}$$

Constraints (5)–(7) select one common battery option for all buses assigned to each route and define the induced route battery capacity $B_{\omega,r}$. With representative-day equivalent capacity cost c^{bat}, the scenario battery cost is

$$C^{\text{bat}}_{\omega}(z_\omega) = c^{\text{bat}} \sum_{r\in\mathcal{R}} n_{\omega,r} \sum_{k\in\mathcal{K}} \beta_k z_{\omega,r,k}.$$

Shared plug-in charging.. Plug-in charger use is modeled over scenario time slots $\mathcal{T}_\omega$. For route r, $\mathcal{V}^{\text{P}}_{\omega,r} \subseteq \mathcal{V}_{\omega,r}$ contains the visit event copies eligible for plug-in charging. An eligible visit event v can use plug-in charging only during $\mathcal{T}_v \subseteq \mathcal{T}_\omega$, and $s(v)$ maps it to its physical station. Let q^{P}_v denote the grid-side plug-in energy supplied during visit v. The binary variable $x_{v\tau}$ records whether visit v occupies a plug-in charger in time slot τ; each occupied slot delivers up to κ units of grid-side energy, and station s has capacity U_s. The plug-in charging constraints are

$$0 \le q^{\text{P}}_v \le \kappa \sum_{\tau\in\mathcal{T}_v} x_{v\tau}, \qquad r \in \mathcal{R},\ v \in \mathcal{V}^{\text{P}}_{\omega,r}, \tag{8}$$

$$\sum_{r\in\mathcal{R}} \sum_{\substack{v\in\mathcal{V}^{\text{P}}_{\omega,r}:\\ s(v)=s,\ \tau\in\mathcal{T}_v}} x_{v\tau} \le U_s, \qquad s \in C,\ \tau \in \mathcal{T}_\omega, \tag{9}$$

$$x_{v\tau} \in \{0,1\}, \qquad r \in \mathcal{R},\ v \in \mathcal{V}^{\text{P}}_{\omega,r},\ \tau \in \mathcal{T}_v. \tag{10}$$

Constraints (8)–(10) limit energy delivery during each visit and enforce available plug-in capacity at every station–time pair. Constraint (9) provides the main cross-route coupling within a scenario.

Route-local operations and charging cost.. Let y_ω collect the nonnegative continuous charging and energy-state variables q_v^{P}, q_v^{S}, q_a^{D}, q_m^{N}, e_v^{arr}, and e_v^{dep}. For each traversal event a, $\ell(a)$ denotes its physical directed link, while tail(a) and head(a) denote its departure and arrival visit event copies. At daytime and overnight electricity prices $c^{\mathrm{day}} \geq 0$ and $c^{\mathrm{night}} \geq 0$, the scenario charging cost is

$$C_\omega^{\mathrm{chg}}(y_\omega) = c^{\mathrm{day}} \left(\sum_{r \in \mathcal{R}} \sum_{v \in \mathcal{V}_{\omega,r}^{\mathrm{P}}} q_v^{\mathrm{P}} + \sum_{r \in \mathcal{R}} \sum_{v \in \mathcal{V}_{\omega,r}^{\mathrm{S}}} q_v^{\mathrm{S}} + \sum_{r \in \mathcal{R}} \sum_{a \in \mathcal{A}_{\omega,r}} q_a^{\mathrm{D}} \right) + c^{\mathrm{night}} \sum_{r \in \mathcal{R}} \sum_{m \in \mathcal{M}_{\omega,r}} q_m^{\mathrm{N}}.$$

The parameters d_a, d_m^0, and d_m^N represent traversal and depot-boundary energy requirements, while $\underline{\rho}$ and $\overline{\rho}$ are the admissible state-of-charge fractions. Trip-level energy requirements are exogenous and independent of the selected battery capacity; each trip chain starts at the upper state-of-charge limit, and overnight depot charging is sufficient to restore every bus before the next representative day. The route-local operational constraints are

$$0 \leq q_v^{\mathrm{S}} \leq \bar{q}_v^{\mathrm{S}} w_{s(v)}^{\mathrm{S}}, \qquad v \in \mathcal{V}_{\omega,r}^{\mathrm{S}}, \tag{11}$$

$$0 \leq q_a^{\mathrm{D}} \leq \bar{q}_a^{\mathrm{D}} w_{\ell(a)}^{\mathrm{D}}, \qquad a \in \mathcal{A}_{\omega,r}. \tag{12}$$

$$\underline{\rho} B_{\omega,r} \leq e_v^{\mathrm{arr}} \leq \overline{\rho} B_{\omega,r}, \qquad v \in \mathcal{V}_{\omega,r}, \tag{13}$$

$$\underline{\rho} B_{\omega,r} \leq e_v^{\mathrm{dep}} \leq \overline{\rho} B_{\omega,r}, \qquad v \in \mathcal{V}_{\omega,r}, \tag{14}$$

$$e_v^{\mathrm{dep}} = e_v^{\mathrm{arr}} + \eta^{\mathrm{P}} q_v^{\mathrm{P}}, \qquad v \in \mathcal{V}_{\omega,r}^{\mathrm{P}}, \tag{15}$$

$$e_v^{\mathrm{dep}} = e_v^{\mathrm{arr}} + \eta^{\mathrm{S}} q_v^{\mathrm{S}}, \qquad v \in \mathcal{V}_{\omega,r}^{\mathrm{S}}, \tag{16}$$

$$e_{\mathrm{head}(a)}^{\mathrm{arr}} = e_{\mathrm{tail}(a)}^{\mathrm{dep}} - d_a + \eta^{\mathrm{D}} q_a^{\mathrm{D}}, \qquad a \in \mathcal{A}_{\omega,r}, \tag{17}$$

$$e_{v_m^0}^{\mathrm{arr}} = \overline{\rho} B_{\omega,r} - d_m^0, \qquad m \in \mathcal{M}_{\omega,r}, \tag{18}$$

$$e_{v_m^N}^{\mathrm{dep}} - d_m^N \geq \underline{\rho} B_{\omega,r}, \qquad m \in \mathcal{M}_{\omega,r}, \tag{19}$$

$$\eta^{\mathrm{P}} q_m^{\mathrm{N}} = \overline{\rho} B_{\omega,r} - e_{v_m^N}^{\mathrm{dep}} + d_m^N, \qquad m \in \mathcal{M}_{\omega,r}. \tag{20}$$

Constraints (11)–(12) link wireless charging quantities to installed facilities, while constraints (13)–(14) bound the energy states relative to $B_{\omega,r}$. Constraints (15)–(17) propagate energy through charging and traversals, and constraints (18)–(20) close each trip chain.

3.3. Recourse structure

For a fixed facility layout w, the expected recourse term is separable across scenarios. This yields a standard integer L-shaped reference in which each scenario is evaluated as a monolithic mixed-integer recourse problem over $(z_\omega, x_\omega, y_\omega)$; we refer to this direct decomposition as Natural ILS.

Within each scenario, battery and charging costs are additive across routes, and all constraints except constraint (9) involve variables associated with a single route. The latter aggregates plug-in occupancy across routes at each station–time pair. That is, the scenario recourse is conditionally route-separable apart from this localized shared-capacity coupling. The formulation also induces a facility–battery resource order. Because trip-level energy consumption is independent of battery capacity, retaining all installed wireless facilities while adding facilities and/or weakly increasing route battery capacities preserves operational feasibility. Moreover, the minimum charging cost under such a resource-dominating configuration is no greater than that under the original configuration.

The recourse structure therefore combines scenario separability, a route-local block structure with limited shared-capacity coupling, and a facility–battery resource order. Section 4 makes the scenario-dependent battery configuration explicit through an equivalent lifted value-function reformulation. The resulting representation, conditional on (w, z_ω), supports both route-level lower bounding and resource-monotonicity strengthening.

4. Methodology

This section develops an exact integer L-shaped method for the two-stage formulation. The method has three main components: a lifted value-function reformulation, a two-phase separation scheme, and resource-monotonicity-strengthened cuts.

4.1. Lifted value-function reformulation

In a natural integer L-shaped decomposition of (1), only the facility variables w are explicit among the discrete configuration decisions, while each scenario recourse evaluation jointly optimizes the battery configuration z_ω and the operational decisions (x_ω, y_ω). This aggregation keeps the battery choice embedded in the mixed-integer recourse problem. Consequently, the master-side support of the integer L-shaped cuts is restricted to the facility variables.

We therefore make z_ω explicit in the lifted decomposition. The facility layout w remains a here-and-now decision, whereas z_ω and (x_ω, y_ω) remain scenario-adaptive decisions made after scenario ω is realized. Thus, lifting changes which variables are visible to the decomposition master, but it does not change the information structure.

To make this separation explicit, we first condition on a candidate facility–battery configuration. For a fixed (w, z_ω), the route battery capacities are fixed through (6), and the remaining scenario evaluation is the exact operational MIP

$$\begin{aligned} Q_\omega(w, z_\omega) = \min_{x_\omega, y_\omega} \quad & C^{\mathrm{chg}}_\omega(y_\omega) \\ \text{s.t.} \quad & (8)\text{–}(10), \\ & (11)\text{–}(20), \quad r \in \mathcal{R}. \end{aligned} \tag{21}$$

The first constraint group retains scenario-level plug-in occupancy and shared-capacity constraints, while the second contains all route-indexed operational constraints. If this MIP is infeasible, we set $Q_\omega(w, z_\omega) = +\infty$.

With Q_ω defined in this way, the scenario value in (4) satisfies

$$\begin{aligned} \Phi_\omega(w) = \min_{z_\omega} \quad & C^{\mathrm{bat}}_\omega(z_\omega) + Q_\omega(w, z_\omega) \\ \text{s.t.} \quad & (5)\text{–}(7). \end{aligned} \tag{22}$$

Let $z = (z_\omega)_{\omega\in\Omega}$ collect the scenario-dependent battery-selection vectors. Substituting (22) into the original two-stage formulation (1) yields the following exact lifted value-function formulation:

$$\begin{aligned} V^L = \min_{w,z} \quad & \left[C^{\mathrm{fac}}(w) + \sum_{\omega\in\Omega} p_\omega C^{\mathrm{bat}}_\omega(z_\omega) \right] + \sum_{\omega\in\Omega} p_\omega Q_\omega(w, z_\omega) \\ \text{s.t.} \quad & (2)\text{–}(3), \\ & (5)\text{–}(7), \qquad\qquad \omega \in \Omega. \end{aligned} \tag{23}$$

In this formulation, both the facility variables w and the scenario-dependent battery variables z_ω are explicit in the decomposition, whereas the remaining operational decisions are represented through the conditional value function $Q_\omega(w, z_\omega)$. The lifted formulation is exact because (22) simply rewrites each scenario recourse value as the sum of battery cost and the exact conditional operational value. It changes the decomposition structure but not the decision timing, information structure, feasible first-stage decisions, or optimal value of (1).

4.2. Relaxation and route-level separation

The lifted reformulation makes the scenario battery variables explicit, but exact lifted integer L-shaped separation remains computationally demanding. For each candidate (w, z_ω), exact scenario evaluation must solve an operational MIP with route coupling from shared plug-in capacity and binary plug-in occupancy. The route-level lower-bounding procedure developed in this section constitutes Phase 1 of the proposed method.

Partition the operating vector by route as $y_\omega = (y_{\omega,r})_{r\in\mathcal{R}}$, with $C^{\mathrm{chg}}_\omega(y_\omega) = \sum_{r\in\mathcal{R}} C^{\mathrm{chg}}_{\omega,r}(y_{\omega,r})$. The plug-in charging block (8)–(10) comprises charging-limit, shared-capacity, and occupancy-domain constraint families. Phase 1 removes the shared station–time capacity family (9) and equivalently replaces the other two families with

$$0 \le q^{\mathrm{P}}_v \le \kappa |\mathcal{T}_v|, \qquad v \in \mathcal{V}^{\mathrm{P}}_{\omega,r}. \tag{24}$$

The resulting relaxation separates by route. For a fixed route configuration $(w, z_{\omega,r})$, all constraints in the following LP are imposed for route r. The route-level LP value is

$$\begin{aligned} Q^R_{\omega,r}(w, z_{\omega,r}) = \min_{y_{\omega,r}} \quad & C^{\mathrm{chg}}_{\omega,r}(y_{\omega,r}) \\ \text{s.t.} \quad & (11)\text{–}(20), \\ & (24). \end{aligned} \tag{25}$$

We set $Q^R_{\omega,r}(w, z_{\omega,r}) = +\infty$ if this route LP is infeasible. Let $Q^R_\omega(w, z_\omega) = \sum_{r\in\mathcal{R}} Q^R_{\omega,r}(w, z_{\omega,r})$ denote the aggregate Phase 1 value for scenario ω. Substituting Q^R_ω for Q_ω in (23) gives the Phase 1 lower-bounding formulation:

$$\begin{aligned} V^R = \min_{w,z} \quad & \left[C^{\text{fac}}(w) + \sum_{\omega\in\Omega} p_\omega C^{\text{bat}}_\omega(z_\omega)\right] + \sum_{\omega\in\Omega} p_\omega Q^R_\omega(w, z_\omega) \\ \text{s.t.} \quad & (2)\text{–}(3), \\ & (5)\text{–}(7), \qquad\qquad \omega \in \Omega. \end{aligned} \tag{26}$$

Proposition 1 (Validity of the Phase 1 relaxation)**.** *For any lifted design configuration (w, z_ω) satisfying the design and battery-selection constraints,*

$$Q^R_\omega(w, z_\omega) \le Q_\omega(w, z_\omega). \tag{27}$$

If any route-level Phase 1 LP is infeasible, the exact operational subproblem is infeasible.

Proof. If the exact operational problem is infeasible, then $Q_\omega(w, z_\omega) = +\infty$ and (27) holds by convention. Otherwise, consider any feasible exact operational solution (x_ω, y_ω). Constraints (8) and (10) imply that its plug-in charging amounts satisfy (24). All remaining constraints are unchanged and route-local; hence, the route-specific components $y_{\omega,r}$ form a feasible solution to (25) for every route r. Because the charging cost is route additive, the aggregate cost of these route solutions equals $C^{\text{chg}}_\omega(y_\omega)$. Minimizing over the route LPs and then over the exact operational solutions gives $Q^R_\omega(w, z_\omega) \le Q_\omega(w, z_\omega)$. The same construction shows that exact feasibility requires every route LP to be feasible, proving the final statement. □

The converse does not hold in general: feasibility of all route LPs does not guarantee exact scenario feasibility because Phase 1 omits the shared plug-in capacity constraints. Because (26) and (23) have the same lifted design domain, Proposition 1 implies $V^R \le V^L$.

To solve the Phase 1 formulation by integer L-shaped separation, introduce an epigraph proxy $\theta^R_{\omega,r}$ for each route value $Q^R_{\omega,r}(w, z_{\omega,r})$. Let $\mathcal{F}^R_{\omega,r}$ and $\mathcal{O}^R_{\omega,r}$ denote the current route-level feasibility and optimality cut pools, and write $\theta^R = \{\theta^R_{\omega,r}\}_{\omega\in\Omega, r\in\mathcal{R}}$. The Phase 1 restricted master is

$$\begin{aligned} \min_{w,z,\theta^R} \quad & \left[C^{\text{fac}}(w) + \sum_{\omega\in\Omega} p_\omega C^{\text{bat}}_\omega(z_\omega)\right] + \sum_{\omega\in\Omega} p_\omega \sum_{r\in\mathcal{R}} \theta^R_{\omega,r} \\ \text{s.t.} \quad & (2)\text{–}(3), \\ & (5)\text{–}(7), && \omega \in \Omega, \\ & \theta^R_{\omega,r} \ge 0, && \omega \in \Omega,\ r \in \mathcal{R}, \\ & \psi^R_{\omega,r}(w, z_{\omega,r}) \ge 0, && \omega \in \Omega,\ r \in \mathcal{R},\ \psi^R_{\omega,r} \in \mathcal{F}^R_{\omega,r}, \\ & \theta^R_{\omega,r} \ge \varphi^R_{\omega,r}(w, z_{\omega,r}), && \omega \in \Omega,\ r \in \mathcal{R},\ \varphi^R_{\omega,r} \in \mathcal{O}^R_{\omega,r}. \end{aligned} \tag{28}$$

Each feasible route LP yields a dual optimality cut in $(w, z_{\omega,r}, \theta^R_{\omega,r})$, whereas an infeasible route LP yields a Farkas feasibility cut. Appendix Appendix B.1 gives their explicit forms.

By Proposition 1, Phase 2 can retain the Phase 1 lower bound, Phase 1 route proxies and accumulated Phase 1 cut pools.

4.3. Resource monotonicity and strengthened scenario cuts

We first establish a scenario-level resource order and then use it to strengthen the pointwise integer L-shaped cuts employed in exact scenario separation. As in (21), set $Q_\omega(w, z_\omega) = +\infty$ when the exact operational MIP is infeasible. The argument uses the model specification that trip-level energy requirements do not vary with the selected battery capacity.

Resource order and monotonicity. Let $\mathcal{W}$ index all candidate SWC stations and DWC links, and let w_j denote the corresponding binary facility variable. Fixed plug-in charger capacities are not included in this resource order. For route r, write $B_{\omega,r}(z_{\omega,r}) = \sum_{k\in\mathcal{K}} \beta_k z_{\omega,r,k}$. For two scenario configurations satisfying the design constraints (2)–(3) and battery-selection constraints (5)–(7), write $(w, z_\omega) \preceq_\omega (\bar{w}, \bar{z}_\omega)$ if

$$w_j \le \bar{w}_j, \quad j \in \mathcal{W}, \qquad B_{\omega,r}(z_{\omega,r}) \le B_{\omega,r}(\bar{z}_{\omega,r}), \quad r \in \mathcal{R}.$$

Proposition 2 (Scenario-level resource monotonicity)**.** *For a fixed scenario $\omega \in \Omega$, if $(w, z_\omega) \preceq_\omega (\bar{w}, \bar{z}_\omega)$, then*

$$Q_\omega(\bar{w}, \bar{z}_\omega) \le Q_\omega(w, z_\omega).$$

Proof. If $Q_\omega(w, z_\omega) = +\infty$, the inequality is immediate. Otherwise, take any feasible exact operating solution (x_ω, y_ω) under (w, z_ω) and keep all occupancy and charging decisions unchanged. The plug-in constraints remain feasible, and the SWC and DWC charging bounds remain valid because the corresponding facility variables weakly increase. For each route r, write $B_{\omega,r} = B_{\omega,r}(z_{\omega,r})$, $\bar{B}_{\omega,r} = B_{\omega,r}(\bar{z}_{\omega,r})$, and $\Delta_r = \bar{B}_{\omega,r} - B_{\omega,r} \geq 0$. Increase every arrival and departure energy state on that route by $\overline{\rho}\Delta_r$. This common shift preserves all station and link energy-balance equations. Moreover, if $\underline{\rho} B_{\omega,r} \leq e \leq \overline{\rho} B_{\omega,r}$, then, since $\overline{\rho} \geq \underline{\rho}$,

$$\underline{\rho}\bar{B}_{\omega,r} \leq e + \overline{\rho}\Delta_r \leq \overline{\rho}\bar{B}_{\omega,r}.$$

The initial boundary equality is preserved exactly, while the return lower bound remains valid because its left-hand side increases by $\overline{\rho}\Delta_r$ and its right-hand side by $\underline{\rho}\Delta_r$. It also preserves the overnight balance because both $\overline{\rho}B_{\omega,r}$ and the terminal departure energy increase by $\overline{\rho}\Delta_r$.

Hence the transformed solution is feasible under $(\bar{w}, \bar{z}_\omega)$ and has the same operational objective value. Minimization therefore gives $Q_\omega(\bar{w}, \bar{z}_\omega) \leq Q_\omega(w, z_\omega)$. □

This monotonicity concerns Q_ω only and does not imply monotonicity of $C^{\text{bat}}_\omega(z_\omega) + Q_\omega(w, z_\omega)$, because larger batteries may have higher battery cost.

Corollary 1 (Scenario-level dominance implications)**.** *For any two scenario configurations* (w, z_ω) *and* (w^*, z^*_ω) *satisfying the design and battery-selection constraints, Proposition 2 implies:*

$$Q_\omega(w^*, z^*_\omega) = +\infty,\ (w, z_\omega) \preceq_\omega (w^*, z^*_\omega) \Rightarrow Q_\omega(w, z_\omega) = +\infty, \tag{29}$$

$$Q_\omega(w^*, z^*_\omega) < +\infty,\ (w, z_\omega) \preceq_\omega (w^*, z^*_\omega) \Rightarrow Q_\omega(w, z_\omega) \geq Q_\omega(w^*, z^*_\omega). \tag{30}$$

Pointwise and strengthened cuts. Consider an evaluated scenario configuration (w^*, z^*_ω). Let $Q^*_\omega = Q_\omega(w^*, z^*_\omega)$ denote its exact operational value, and let $k^*_{\omega,r}$ be the selected battery option on route r. Because charging costs are nonnegative, we use $L_\omega = 0$ as a global lower bound and impose $\theta_\omega \geq L_\omega$. The pointwise mismatch expression is

$$D_\omega(w, z_\omega; w^*, z^*_\omega) = \sum_{j \in \mathcal{W}: w^*_j = 0} w_j + \sum_{j \in \mathcal{W}: w^*_j = 1} (1 - w_j) + \sum_{r \in \mathcal{R}} \left(1 - z_{\omega,r,k^*_{\omega,r}}\right).$$

The pointwise scenario cuts are

$$\begin{aligned} D_\omega(w, z_\omega; w^*, z^*_\omega) &\geq 1, && \text{if } Q^*_\omega = +\infty, \\ \theta_\omega &\geq Q^*_\omega - (Q^*_\omega - L_\omega) D_\omega(w, z_\omega; w^*, z^*_\omega), && \text{if } Q^*_\omega < +\infty. \end{aligned}$$

Under the binary facility domains and the route-level battery-choice constraints, $D_\omega = 0$ if and only if $(w, z_\omega) = (w^*, z^*_\omega)$; otherwise $D_\omega \geq 1$, yielding the standard pointwise feasibility and optimality corrections.

Corollary 1 extends these pointwise corrections to dominated resource configurations. Define the scenario-level monotonicity mismatch expression

$$I_\omega(w, z_\omega; w^*, z^*_\omega) = \sum_{j \in \mathcal{W}: w^*_j = 0} w_j + \sum_{r \in \mathcal{R}} \sum_{k \in \mathcal{K}: \beta_k > \beta_{k^*_{\omega,r}}} z_{\omega,r,k}.$$

The monotonicity-strengthened scenario cuts are

$$\begin{aligned} I_\omega(w, z_\omega; w^*, z^*_\omega) &\geq 1, && \text{if } Q^*_\omega = +\infty, \\ \theta_\omega &\geq Q^*_\omega - (Q^*_\omega - L_\omega) I_\omega(w, z_\omega; w^*, z^*_\omega), && \text{if } Q^*_\omega < +\infty. \end{aligned}$$

Under the binary facility domains and the route-level battery-choice constraints, I_ω is a nonnegative integer. Moreover, $I_\omega = 0$ implies $(w, z_\omega) \preceq_\omega (w^*, z^*_\omega)$. If $Q^*_\omega = +\infty$, Corollary 1 therefore implies that every configuration with $I_\omega = 0$ is infeasible, proving the strengthened feasibility cut. If $Q^*_\omega < +\infty$ and $I_\omega = 0$, the same corollary gives $Q_\omega(w, z_\omega) \geq Q^*_\omega$. For $I_\omega \geq 1$, the feasibility cut is satisfied and

$$Q^*_\omega - (Q^*_\omega - L_\omega) I_\omega \leq L_\omega,$$

so the optimality cut is valid under $\theta_\omega \geq L_\omega$. Since $I_\omega \leq D_\omega$ and $Q^*_\omega - L_\omega \geq 0$, the strengthened feasibility and optimality cuts weakly dominate their pointwise counterparts.

4.4. Complete procedure and convergence

Let LB^R and UB^R denote the current lower and upper bounds on the Phase 1 formulation (26). For a prescribed phase-switch tolerance $\varepsilon_1 \geq 0$, Phase 1 switches when

$$g^R = \frac{UB^R - LB^R}{\max\{1, |UB^R|\}} \leq \varepsilon_1.$$

Set $g^R = +\infty$ until a finite UB^R is available. By Proposition 1, LB^R is a valid lower bound on both the Phase 1 formulation and the exact lifted formulation, whereas UB^R applies only to the Phase 1 relaxation and is not an exact incumbent bound. The tolerance ε_1 controls the computational effort devoted to Phase 1 without affecting the validity of Phase 2.

At the phase transition, Phase 2 retains the Phase 1 route proxies and route-level cut pools and introduces a scenario-level proxy θ_ω for the exact operational value $Q_\omega(w, z_\omega)$. The linking constraint $\theta_\omega \geq \sum_{r\in\mathcal{R}} \theta^R_{\omega,r}$ transfers the retained route-level lower bounds to the scenario level. Let $\mathcal{F}_\omega$ denote the Phase 2 scenario-level feasibility cut pool, including RSP and exact scenario feasibility cuts, and let O_ω denote the corresponding optimality cut pool. The Phase 2 restricted master is

$$\begin{aligned}
\min_{w,z,\theta^R,\theta} \quad & \left[C^{\text{fac}}(w) + \sum_{\omega\in\Omega} p_\omega C^{\text{bat}}_\omega(z_\omega)\right] + \sum_{\omega\in\Omega} p_\omega \theta_\omega \\
\text{s.t.} \quad & (2)\text{–}(3), \\
& (5)\text{–}(7), && \omega \in \Omega, \\
& \theta^R_{\omega,r} \geq 0, && \omega \in \Omega,\ r \in \mathcal{R}, \\
& \psi^R_{\omega,r}(w, z_{\omega,r}) \geq 0, && \omega \in \Omega,\ r \in \mathcal{R},\ \psi^R_{\omega,r} \in \mathcal{F}^R_{\omega,r}, \\
& \theta^R_{\omega,r} \geq \varphi^R_{\omega,r}(w, z_{\omega,r}), && \omega \in \Omega,\ r \in \mathcal{R},\ \varphi^R_{\omega,r} \in O^R_{\omega,r}, \\
& \theta_\omega \geq \sum_{r\in\mathcal{R}} \theta^R_{\omega,r}, && \omega \in \Omega, \\
& \theta_\omega \geq L_\omega, && \omega \in \Omega, \\
& \psi_\omega(w, z_\omega) \geq 0, && \omega \in \Omega,\ \psi_\omega \in \mathcal{F}_\omega, \\
& \theta_\omega \geq \varphi_\omega(w, z_\omega), && \omega \in \Omega,\ \varphi_\omega \in O_\omega.
\end{aligned} \tag{31}$$

Before invoking exact MIP separation, each Phase 2 master candidate (w, z, θ^R, θ) is screened scenario by scenario using a scenario-level relaxed subproblem (RSP). For each feasible RSP, violated RSP optimality cuts are added, whereas an infeasible RSP certifies infeasibility of the corresponding exact operational MIP and yields a valid scenario-level feasibility cut before exact separation. When the RSPs are feasible, the algorithm proceeds to exact scenario separation.

If the exact operational MIP is feasible and solved to optimality for every scenario at a restricted-master candidate (w, z), the corresponding feasible objective value is

$$UB^{\text{cand}}(w, z) = C^{\text{fac}}(w) + \sum_{\omega\in\Omega} p_\omega C^{\text{bat}}_\omega(z_\omega) + \sum_{\omega\in\Omega} p_\omega Q_\omega(w, z_\omega). \tag{32}$$

The incumbent is updated as $UB \leftarrow \min\{UB, UB^{\text{cand}}(w, z)\}$, and the optimal value of the current Phase 2 restricted master is a valid lower bound LB. For a finite incumbent UB, define the gap as $g = (UB - LB)/\max\{1, |UB|\}$. Set $g = +\infty$ until a finite incumbent is available. Algorithm 1 summarizes the complete procedure for a final tolerance $\varepsilon \geq 0$.

Theorem 1 (Finite convergence under exact separation)**.** *Assume that $p_\omega > 0$ for all $\omega \in \Omega$, the lifted formulation has a finite optimum, and the binary facility–battery configuration space is finite. Suppose Phase 1 switches after finitely many iterations, all generated cuts are valid and retained, auxiliary RSP separation terminates finitely at each master candidate, and every restricted master and exact Phase 2 scenario MIP is solved to proven infeasibility or optimality. With $\varepsilon = 0$, Algorithm 1 terminates finitely with an optimal solution to the lifted formulation and hence to the original two-stage formulation.*

The proof is provided in Appendix Appendix B.

For $\varepsilon > 0$, the procedure returns the incumbent and valid bounds once $g \leq \varepsilon$; if no finite incumbent has been obtained, $g = +\infty$ and gap-based termination is not declared.

Input : Scenario data, candidate facilities and batteries, phase-switch tolerance ε_1, final tolerance ε
Output: Facility layout w, scenario battery configurations z, lower bound LB, upper bound UB

Initialize cut pools; set $LB^R, LB \leftarrow -\infty$ and $UB^R, UB \leftarrow +\infty$;

`// Phase 1: route-level lower-bounding separation`
repeat
  Solve (28) to optimality, set LB^R to its optimal value, and update $LB \leftarrow \max\{LB, LB^R\}$;
  Solve route-level LP relaxations for each scenario at the current (w, z_ω);
  Add violated route-level dual optimality cuts for feasible route LPs;
  if *one or more route LPs are infeasible* **then**
    Add violated route-level Farkas cuts and continue;
  end
  Set $UB^R \leftarrow \min\{UB^R,$ evaluated Phase 1 candidate value$\}$;
until $g^R \le \varepsilon_1$;
Transfer LB, route proxies, and route cut pools; initialize (31);

`// Phase 2: exact mixed-integer verification`
repeat
  Solve the restricted master (31) to optimality and set LB to its optimal value;
  Solve the relaxed subproblems (RSPs) for all scenarios;
  Add violated RSP optimality cuts for feasible RSPs;
  if *one or more RSPs are infeasible* **then**
    Add the corresponding RSP feasibility cuts;
    continue to the next master solve;
  end
  Solve the exact scenario operational MIPs to proven infeasibility or optimality;
  if *one or more scenarios are proven infeasible* **then**
    Add the pointwise or strengthened scenario feasibility cuts defined above;
    continue to the next master solve;
  end
  if *all scenario MIPs are proven optimal* **then**
    Compute UB^{cand} using (32);
    Set $UB \leftarrow \min\{UB, UB^{\text{cand}}\}$ and update the incumbent if improved;
    foreach *scenario whose exact recourse is underestimated by* θ_ω **do**
      Add the pointwise or strengthened scenario optimality cuts defined above;
    end
  end
until $g \le \varepsilon$;
return *Best incumbent, LB, and UB*;

Algorithm 1: Two-phase integer L-shaped method

5. Computational Experiments

This section evaluates the proposed method on instances derived from an operating bus network. We first define the base instance and characterize a representative solution, then evaluate the algorithmic components and scalability, and finally examine sensitivity to problem characteristics.

5.1. Experimental setup and base instance

The experiments use a study network based on the Tower Transit bus network in Singapore. The network comprises 28 routes, 493 physical stations, 625 directed links, and five existing plug-in charging locations, as shown in Figure 3. The interacting routes and shared charging resources induce the cross-route coupling represented in the operational model.

The base service plan is constructed from official Tower Transit route and timetable data and contains 282 buses and 2,850 scheduled trips. Two additional service scenarios are generated by decreasing and increasing systemwide route frequencies by 10%, with trip chains and fleet requirements recomputed accordingly. The three scenarios are equally likely. Other physical and economic parameters and their sources are reported in Appendix Appendix C.1.

The algorithms are implemented in C++ using Gurobi 12.0.3 and run on an Intel Core i5-14600K processor with 32 GB of memory under a 12-thread budget. Each run has a 7,200-second time limit and uses the outer termination tolerance $\varepsilon = 10^{-4}$. Our code and data are available in the GitHub repository (`https://github.com/foangfangkeeke/Two_phase_for_IJOC`).

5.2. Representative planning solution

We use the base instance to characterize a representative solution and compare it with a controlled benchmark without WPT. The benchmark disables SWC and DWC while retaining the same service scenarios and plug-in charging resources and reoptimizing the scenario-dependent battery configurations and charging operations.

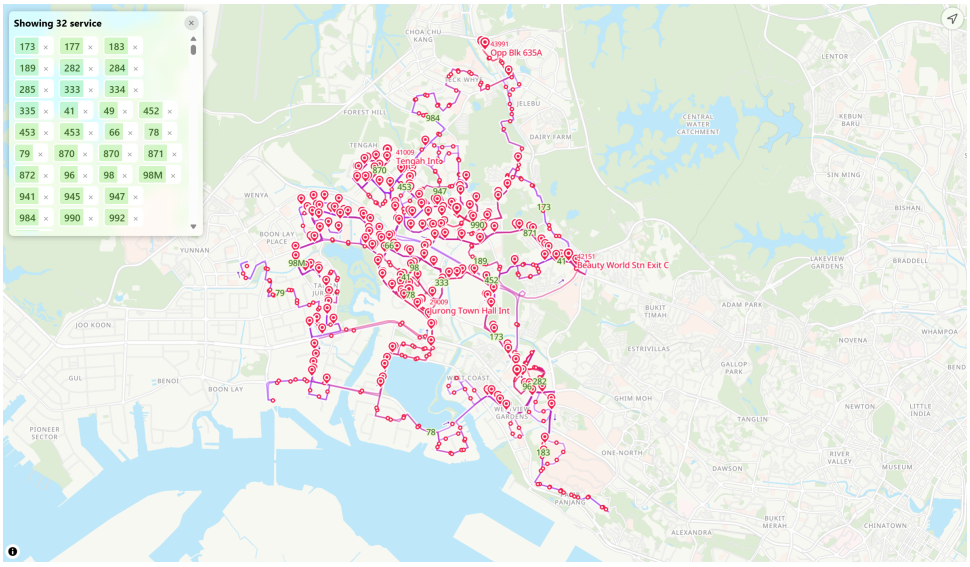


Figure 3: Spatial distribution of the Tower Transit bus network used in the case study.

This comparison isolates the effect of allowing WPT investment on the joint infrastructure, battery, and charging decisions.

Relative to the no-WPT benchmark displayed in Table 1, the WPT solution adds \$192.240 per day in facility cost and \$917.451 per day in charging cost while reducing battery cost by \$3,327.040 per day. The net reduction is \$2,217.349 per day, or 21.98%. The comparison indicates that the model substitutes wireless-facility investment and additional charging expenditure for substantially lower battery expenditure.

Table 1: Objective components with and without wireless power transfer

	Cost (USD/day)	
Component	Without WPT	With WPT
Wireless facilities	0.000	192.240
Scenario-dependent batteries	4,183.467	856.427
Charging operations	5,903.714	6,821.165
Total	10,087.181	7,869.832

Note. The reported components exclude scenario-dependent constants omitted from the optimization formulation.

The battery configurations reflect the same substitution. With WPT, all 28 routes choose the 20-kWh option in every scenario; without WPT, only eight routes select the 20-kWh option in each scenario, the mean route battery capacity ranges from 82.86 to 85.71 kWh, and the maximum is 310 kWh. The representative plan selects 84 of the 488 candidate SWC stations and two of the 625 candidate DWC links. Table 2 compares the average operational exposure of all candidates with that of the selected facilities within each charging mode and reports the average number of charging events at each selected facility.

Table 2: Operational exposure and charging events at wireless facilities

Mode	Scenario	Avg. events per candidate facility	Avg. events per selected facility	Avg. charging events per selected facility
SWC	Low	138.88	212.88	125.31
SWC	Base	154.18	236.31	139.80
SWC	High	169.80	260.30	157.26
DWC	Low	112.94	187.50	23.50
DWC	Base	125.40	209.00	27.00
DWC	High	138.09	229.00	30.50

Note. Events are scheduled-trip arrivals for SWC and directed-traversal event copies for DWC; charging events count positive charging decisions.

Across all three scenarios, selected SWC stations and selected DWC links have higher average operational exposure than their respective candidate sets, and positive charging events are observed for both selected facility types. This pattern is consistent with WPT investment being concentrated at relatively high-exposure locations. Overall, the benchmark comparison shows that allowing WPT produces a nontrivial joint planning response: the additional infrastructure and charging expenditures are offset by substantially lower battery expenditure, while the selected wireless facilities are actively used across the service scenarios.

5.3. Algorithmic performance and scalability

This section first uses an ablation comparison to identify the contribution of the proposed algorithmic components and then examines computational performance as network size and scenario count increase. We denote an instance by *N*-*k*, where $N \in \{\mathrm{S}, \mathrm{M}, \mathrm{B}, \mathrm{L}\}$ indicates increasing network size, with B denoting the base network mentioned in Section 5.1, and k is the number of service-plan scenarios. For example, B-3 represents the instance used in Section 5.2.

We use EF for the extensive-form formulation and ILS for the natural integer L-shaped method. In the remaining method names, L denotes the lifted reformulation, M denotes the monotonicity-strengthened scenario cuts, and 2P denotes the two-phase separation scheme. The resulting variants are EF, ILS, L-ILS, L-ILS-M, L2P-ILS, and Full, where Full combines the lifted reformulation, two-phase separation, and monotonicity strengthening. In particular, ILS versus L-ILS isolates lifting, while the remaining matched pairs identify the effects of two-phase separation and monotonicity strengthening under otherwise identical configurations. All decomposition variants use RSP screening, which is held fixed rather than treated as an ablation component.

Table 3 summarizes the component configuration of each method.

Table 3: Algorithmic components included in each method

Method	Integer L-shaped decomposition	Lifted reformulation	Monotonicity-strengthened cuts	Two-phase separation
EF	×	×	×	×
ILS	✓	×	×	×
L-ILS	✓	✓	×	×
L-ILS-M	✓	✓	✓	×
L2P-ILS	✓	✓	×	✓
Full	✓	✓	✓	✓

Note. A check mark indicates that the named component is included.

The matched comparisons in Table 5 report the performance consequences of these component configurations as network size increases.

5.3.1. Network scalability

The network-scale experiment fixes three service-plan scenarios and uses Table 4 to define S-3, M-3, B-3, and L-3 through growth in routes, trips, duties, and candidate facilities.

Table 4: Dimensions of the network-scale instances

Instance	Routes	Stations	Links	Trips	Duties	SWC candidates	DWC candidates	Plug-in locations
S-3	3	49	53	206/230/252	26/28/29	47	53	2
M-3	19	425	515	623/690/761	69/70/76	420	515	5
B-3	28	493	625	2,568/2,850/3,138	264/282/322	488	625	5
L-3	59	1,459	1,822	5,669/6,291/6,923	705/762/909	1,451	1,822	8

Note. Trip and duty counts are ordered as low-frequency/base/high-frequency scenarios.

The network-scale performance analysis uses Table 5 to compare the available methods, distinguish unrun methods from failed runs, and identify the observed computational boundary.

5.3.2. Scenario-count scalability

The scenario-scale experiment fixes the base network and defines B-3, B-5, B-10, B-20, and B-40 through their scenario-generation and probability rules.

The scenario-scale performance analysis uses Table 6 to compare the methods for which runs are available as the number of scenarios increases.

The interpretation characterizes the observed runtime and gap progression and identifies the tested solved range without extrapolating asymptotic behavior from five deterministic runs.

Table 5: Computational performance across network scales

Instance	EF	ILS	L-ILS	L-ILS-M	L2P-ILS	Full
	4.194 / 0.000000	0.880 / 0.000000	0.183 / 0.000000	0.185 / 0.000000	0.324 / 0.000000	0.353 / 0.000000
S-3	Opt. 7,201.313 / 0.270300	Conv. 7,390.550 / 0.801414	Conv. 7,205.003 / –	Conv. 7,222.265 / –	Conv. 23.904 / 0.000000	Conv. 25.678 / 0.000000
M-3	TL 563.880 / –	TL	TL-NI	TL-NI	Conv. 28.594 / 0.000310	Conv. 32.228 / 0.000221
B-3	OOM	–	–	–	Conv. 7,217.538 / 0.092176	Conv. 7,231.923 / 0.092176
L-3	–	–	–	–	TL	TL

Note. Each available result reports total time in seconds / final gap in percent, followed by status. Opt., Conv., TL, OOM, NI, and – denote optimal, outer convergence, time limit, out of memory, no finite incumbent, and not run, respectively.

Table 6: Computational performance as the number of scenarios increases

Instance	Scenarios	EF	L2P-ILS	Full
		563.880 / –	28.594 / 0.000310	32.228 / 0.000221
B-3	3	OOM	Conv. 90.016 / 0.003269	Conv. 57.679 / 0.003359
B-5	5	–	Conv. ≥168.504 / 0.222105	Conv. 119.665 / 0.002699
B-10	10	–	IL	Conv. 247.376 / 0.002187
B-20	20	–	–	Conv. 926.361 / 0.002187
B-40	40	–	–	Conv.

Note. Each available result reports total time in seconds / final gap in percent, followed by status. IL denotes an incomplete log; the reported B-10 L2P-ILS time is therefore a lower bound.

5.4. Sensitivity to problem characteristics

The sensitivity experiment fixes the 28-route, ten-scenario instance and varies chargers per plug-in location from two to six, changing exact station–time capacity while leaving the Phase 1 route-level relaxation unchanged.

The transition analysis compares the Phase 1 lower bound LB^R with the first finite exact-feasible Phase 2 objective UB_0 through $g_0 = (UB_0 - LB^R)/\max\{1, |UB_0|\}$ and reports the associated metrics in Table 7. Here, UB_0 evaluates the first reoptimized Phase 2 master candidate rather than the stored Phase 1 incumbent.

Table 7: Shared-capacity sensitivity and Phase 1 lower-bound quality

U_s	LB^R	UB_0	g_0 (%)	Final gap (%)	Phase 1 (s)	Phase 2 (s)	Total (s)	Status
2	7,833.727845	–	–	–	59.963	7,183.166	7,243.129	TL-NI
3	7,833.737000	7,887.027351	0.675671	0.007599	98.517	181.974	280.491	Conv.
4	7,833.737000	7,878.012357	0.562012	0.002699	60.222	87.936	148.158	Conv.
5	7,833.737000	7,875.672628	0.532470	0.002124	62.882	33.008	95.890	Conv.
6	7,833.737000	7,875.265535	0.527329	0.001358	61.027	52.966	113.993	Conv.

Note. U_s is the number of plug-in chargers available at each existing location. The capacity-2 run reaches the time limit without a finite exact-feasible incumbent; its Phase 2 time equals total time minus Phase 1 time.

The results discussion assesses how shared capacity affects the initial Phase 2 gap and phase-specific time and treats the capacity-2 run as a boundary case.

6. Conclusion

This paper studies wireless charging infrastructure planning for electric-bus systems under uncertain future service plans. The two-stage formulation separates here-and-now SWC and DWC deployment from scenario-adaptive route battery configuration and representative-day charging decisions. Its recourse combines scenario-dependent discrete configurations, route-level energy feasibility, and shared plug-in charger capacity. To exploit this structure, we develop an exact integer L-shaped method that lifts the battery configurations into the decomposition master, uses route-level LP recourse to generate lower bounds before exact scenario verification, and strengthens scenario cuts through a joint facility–battery resource order. The resulting procedure preserves the original information structure and converges finitely under exact separation.

The Tower-derived experiments identify the two-phase relaxation–verification scheme as the main computational improvement in the tested method ladder. The two-phase variants solve the small, medium, and base three-scenario instances, whereas the standard integer L-shaped variants do not solve even the small instance within the two-hour limit. On the base network, the full method solves all cases from 3 to 40 scenarios; the 40-scenario instance terminates in 926.4 seconds with a final gap below 0.004%. The boundary cases also clarify where the method remains expensive: the largest three-scenario network reaches the time limit at a 0.092% gap, and the tightest plug-in capacity setting fails to complete exact Phase 2 verification. Thus, after route-level lower bounding reduces the cost of early separation, exact verification becomes the principal bottleneck when the network is large or shared-capacity coupling is tight.

The study uses constructed service-plan scenarios on a fixed route network, treats existing plug-in charger capacity as exogenous, and assumes that trip energy consumption is independent of the selected battery capacity. Future work could develop data-driven service-plan scenarios, jointly optimize wireless and plug-in capacity expansion, and incorporate battery-weight effects and short-term operational uncertainty. From an algorithmic perspective, stronger exact-verification procedures for tightly coupled recourse and tests on other stochastic design problems would help establish how broadly the layered decomposition applies.

Appendix A. Model Notation

Table A.8: Summary of notation

Symbol	Description
Sets and indices	
Ω	Set of service-plan scenarios, indexed by ω
$\mathcal{S}$	Set of stations, indexed by s
$\mathcal{L}$	Set of directed links between consecutive stations, indexed by l
C	Set of existing interchanges or hubs equipped with plug-in chargers, $C \subseteq \mathcal{S}$
$\mathcal{R}$	Set of routes, indexed by r
$\mathcal{K}$	Set of candidate battery-capacity options, indexed by k
$\mathcal{M}_{\omega,r}$	Set of bus trip chains on route r under scenario ω, indexed by m
$\mathcal{V}_{\omega,r}$	Set of station-visit event copies generated by all trip chains on route r under scenario ω, indexed by v
$\mathcal{V}^{\mathrm{P}}_{\omega,r}$	Set of visit event copies eligible for plug-in charging on route r under scenario ω
$\mathcal{V}^{\mathrm{S}}_{\omega,r}$	Set of visit event copies eligible for SWC on route r under scenario ω
$\mathcal{A}_{\omega,r}$	Set of directed-traversal event copies generated by all trip chains on route r under scenario ω, indexed by a
$\mathcal{T}_{\omega}$	Relevant plug-in charging time periods under scenario ω
$\mathcal{T}_{v}$	Time periods during which plug-in charging visit v can occupy a plug-in charger
Parameters	
p_{ω}	Probability of scenario ω
c^{S}_{s}	Representative-day equivalent cost of installing SWC at station s
c^{D}_{l}	Representative-day equivalent cost of installing DWC on link l
c^{bat}	Representative-day equivalent battery capacity cost per kWh
$c^{\mathrm{day}}, c^{\mathrm{night}}$	Daytime and overnight electricity prices
$n_{\omega,r}$	Fleet-size requirement for route r under scenario ω
β_k	Battery capacity value represented by option k
d_a	Energy consumption associated with directed link traversal a
d^0_m	Boundary energy consumption from the depot to the first station visit of trip chain m
d^N_m	Boundary energy consumption from the terminal station visit of trip chain m back to the depot
U_s	Number of existing plug-in chargers at station s
κ	Grid-side energy delivered by one occupied plug-in time slot
$\bar{q}^{\mathrm{S}}_v$	Maximum SWC energy available during station visit v
$\bar{q}^{\mathrm{D}}_a$	Maximum DWC energy available during directed link traversal a
$\eta^{\mathrm{P}}, \eta^{\mathrm{S}}, \eta^{\mathrm{D}}$	Charging efficiencies for plug-in charging, SWC, and DWC
$\underline{\rho}, \bar{\rho}$	Minimum and maximum state-of-charge fractions
Derived quantities and functions	
$s(v)$	Physical station associated with visit event v
$\ell(a)$	Physical directed link associated with traversal event a
tail(a)	Departure visit event of traversal event a
head(a)	Arrival visit event of traversal event a
v^0_m	First visit event of trip chain m
v^N_m	Last visit event of trip chain m
Decision variables	
w^{S}_s	1 if SWC is installed at station $s \in \mathcal{S} \setminus C$, 0 otherwise
w^{D}_l	1 if DWC is installed on link $l \in \mathcal{L}$, 0 otherwise
$z_{\omega,r,k}$	1 if route r uses battery capacity option k in scenario ω, 0 otherwise
$B_{\omega,r}$	Selected battery capacity for route r in scenario ω
x_{ω}	Vector of scenario-level plug-in charger occupancy decisions in scenario ω
y_{ω}	Vector of scenario-level charging amounts and battery energy-state decisions in scenario ω
q^{P}_v	Plug-in charging energy assigned during plug-in visit v
q^{S}_v	SWC energy assigned during SWC-eligible station visit v
q^{D}_a	DWC energy assigned to directed link traversal a
q^{N}_m	Overnight charging energy assigned after trip chain m returns to the depot
$e^{\mathrm{arr}}_v, e^{\mathrm{dep}}_v$	Battery energy state at arrival and departure of station visit v
$x_{v\tau}$	1 if plug-in charging visit v occupies one plug-in charger at time τ, 0 otherwise

Appendix B. Proofs

Appendix B.1. Route-Level LP Cuts for Phase 1

For a fixed scenario–route pair (ω, r), express the Phase 1 route LP (25) in the equivalent parametric form

$$Q^R_{\omega,r}(w, z_{\omega,r}) = \min_{y_{\omega,r}\geq 0} \left\{ c^\top_{\omega,r} y_{\omega,r} : A_{\omega,r} y_{\omega,r} \geq b_{\omega,r} + H_{\omega,r} w + G_{\omega,r} z_{\omega,r} \right\}.$$

Here, $c_{\omega,r}$, $A_{\omega,r}$, $b_{\omega,r}$, $H_{\omega,r}$, and $G_{\omega,r}$ collect the corresponding fixed objective and constraint coefficients. If the route LP is feasible, any dual-feasible vector

$$\pi_{\omega,r} \geq 0, \qquad A^\top_{\omega,r} \pi_{\omega,r} \leq c_{\omega,r},$$

gives the route-level optimality cut

$$\theta^R_{\omega,r} \geq \pi^\top_{\omega,r} \left(b_{\omega,r} + H_{\omega,r} w + G_{\omega,r} z_{\omega,r} \right).$$

If the route LP is infeasible at an evaluated configuration $(w^*, z^*_{\omega,r})$, a Farkas ray can be selected from

$$\Sigma^{\mathrm{F}}_{\omega,r} = \left\{ \sigma_{\omega,r} \geq 0 : A^\top_{\omega,r} \sigma_{\omega,r} \leq 0 \right\}$$

such that

$$\hat{\sigma}^\top_{\omega,r} \left(b_{\omega,r} + H_{\omega,r} w^* + G_{\omega,r} z^*_{\omega,r} \right) > 0.$$

Then any configuration with a feasible Phase 1 route relaxation must satisfy

$$\hat{\sigma}^\top_{\omega,r} \left(b_{\omega,r} + H_{\omega,r} w + G_{\omega,r} z_{\omega,r} \right) \leq 0. \tag{B.1}$$

Thus (B.1) is a valid feasibility cut for the Phase 1 relaxation.

Appendix B.2. Finite convergence of the two-phase method

Proof of Theorem 1. Proposition 1 and cut validity make every Phase 2 restricted master a relaxation of the exact lifted formulation, so its optimal value is a valid lower bound. Whenever all exact scenario subproblems are proven feasible and solved to optimality, (32) gives a valid upper bound. At any Phase 2 master configuration, exact separation either excludes an infeasible scenario configuration or enforces its exact recourse value whenever $\theta_\omega < Q_\omega(w, z_\omega)$. Because the binary configuration space and scenario set are finite and all corrections are retained, only finitely many such corrections can occur.

At the final uncorrected master solution, every scenario is exact-feasible. If $\theta_\omega < Q_\omega(w, z_\omega)$ for any scenario, exact separation would add a further correction; hence $\theta_\omega \geq Q_\omega(w, z_\omega)$ at termination. All retained constraints on θ_ω are valid lower bounds on $Q_\omega(w, z_\omega)$, so decreasing any θ_ω above $Q_\omega(w, z_\omega)$ preserves master feasibility. Because $p_\omega > 0$ and the master minimizes θ_ω, no optimal master solution can choose such an excessive value, and therefore $\theta_\omega = Q_\omega(w, z_\omega)$ for every scenario. The master objective then equals the exact candidate value, so $LB = UB$ when $\varepsilon = 0$. Exactness of the lifted reformulation completes the result. □

Appendix C. Additional Computational Results

Appendix C.1. Computational parameter settings

Facility and battery capital costs are annualized and converted to representative-day equivalents to be comparable with daily charging costs. Table C.9 summarizes the common physical and economic parameters.

Table C.9: Parameter settings for the computational experiments

Parameter	Value	Source
Battery and state of charge		
Battery capacity cost c^{bat}	0.148 USD/kWh/day	BloombergNEF (2025); Wang et al. (2024)
State-of-charge bounds $(\underline{\rho}, \overline{\rho})$	$(0.2, 0.8)$	Li et al. (2024)
Charging operation		
Plug-in energy rate κ	2.5 kWh per one-minute slot	Najafi et al. (2025)
SWC opportunity limit $\bar{q}_v^{\text{S}}$	1.0 kWh/min times eligible dwell time	Wang et al. (2024)
DWC opportunity limit $\bar{q}_a^{\text{D}}$	1.0 kWh/min times traversal time	Wang et al. (2024)
Charging efficiencies $(\eta^{\text{P}}, \eta^{\text{S}}, \eta^{\text{D}})$	$(0.90, 0.85, 0.80)$	Li et al. (2024); Sun et al. (2024)
Infrastructure and electricity cost		
Facility amortization horizon T^{fac}	30 years	Wang et al. (2024)
SWC facility cost c_s^{S}	2.16 USD/day	Wang et al. (2024)
DWC facility cost c_l^{D}	9 USD/km/day times link length, plus 0.90 USD/day	Wang et al. (2024)
Electricity prices $(c^{\text{day}}, c^{\text{night}})$	$(0.1922, 0.1233)$ USD/kWh	Singapore Department of Statistics (2026)

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